BUSINESS MODELS & PROJECT TIMELINE

Enterprise Model Options & 8-Week Execution Plan (Teams of 5 Students)

1. Approved Business Models List (8 Options)

Teams must select one of the following 8 approved enterprise models. Each model is designed to support physical subnet separation across dedicated router interfaces without requiring VLANs or Layer 3 Switches.

Business Model	Departmental & Site Breakdown
1. Multi-Branch English Language Center	Headquarters: Administration, Course Management, Central Server Farm. Branches: Classrooms, Teacher Staff Room, Guest Wi-Fi Network.
2. Regional Logistics & Warehousing Company	Main Office: Executive Management, Finance, Customer Service. Warehouse Site: Inventory Control, Logistics Operations, Security Subnet, IT/Server Room.
3. Private Healthcare Clinic Chain	Main Hospital: Medical Staff, Administration, Patient Records Server. Satellite Clinic: Consultation Rooms, Pharmacy Desk, Reception Wi-Fi.
4. E-Commerce & Retail Business	Corporate Office: Digital Marketing, Sales Operations, IT Infrastructure. Fulfillment Center: Order Processing, Packing Stations, Dispatch Logistics.
5. Chain of Boutique Hotels & Resorts	Main Property: Front Desk Operations, Executive Management, Internal IT Services / Server Room. Branch / Annex Building: Guest Rooms Wi-Fi, Facility Management, POS Terminal Subnet.
6. Architecture & Engineering Firm	Headquarters: Design/CAD Team, Executive Management, Project File Server. Site Office: Project Managers, On-site Engineers, Guest Network.
7. Regional Agricultural Export & Cold Chain	Headquarters: Export Sales, Accounting, Central ERP/Order Server. Processing Plant & Cold Storage: Warehouse

	Management, Environmental Monitoring Subnet, Packaging Lines.
8. Independent Co-Working Space Chain	Operations HQ: General Management, Marketing, Membership & Booking Server. Branch Location: Dedicated Office Subnet, Hot-Desk Member Wi-Fi, Reception & Cafe POS.

2. Project Timeline & Milestones (8 Weeks)

Timeline	Milestone Phase	Expected Deliverables & Key Tasks
Week 1	**Business Model Selection & Topology Sketch**	Select company profile, collect requirements, and design initial physical topology diagram.
Week 2	**IP Addressing Scheme & Subnetting (VLSM)**	Calculate IPv4 VLSM subnets; construct complete IP Addressing Table (.xlsx). Receive unique base IP allocation. (IPv6 prefix planning optional — required only for the IPv6 Dual-Stack extra credit tier.)
Week 3	**LAN Base Configuration**	Configure basic switch parameters, physical interface IP addresses on Routers, and host connections.
Week 4	**WAN Connectivity & Checkpoint Review**	Establish WAN links between Head Office and Branch; configure static routes (OSPF optional for extra credit — see Advanced Features). [CHECKPOINT 1 REVIEW: Instructor assessment of topology & connectivity].
Week 5	**Core Network Services Deployment**	Configure and test DHCP, DNS, HTTPS web portal, and Email services on dedicated server nodes.
Week 6	**Security Implementation & Hardening**	Apply password rules, disable Telnet, enable SSH v2, configure exec timeouts, banners, and optional ACL/AAA.
Week 7	**System Integration Testing & Report Drafting**	Perform end-to-end connectivity verification (ping/traceroute), simulation mode tests, and write final

		report.
Week 8	**Final Submission & Oral Presentation**	Submit `.pkt` file, final `.docx`/`.pdf` report, and present 15-minute project demonstration.